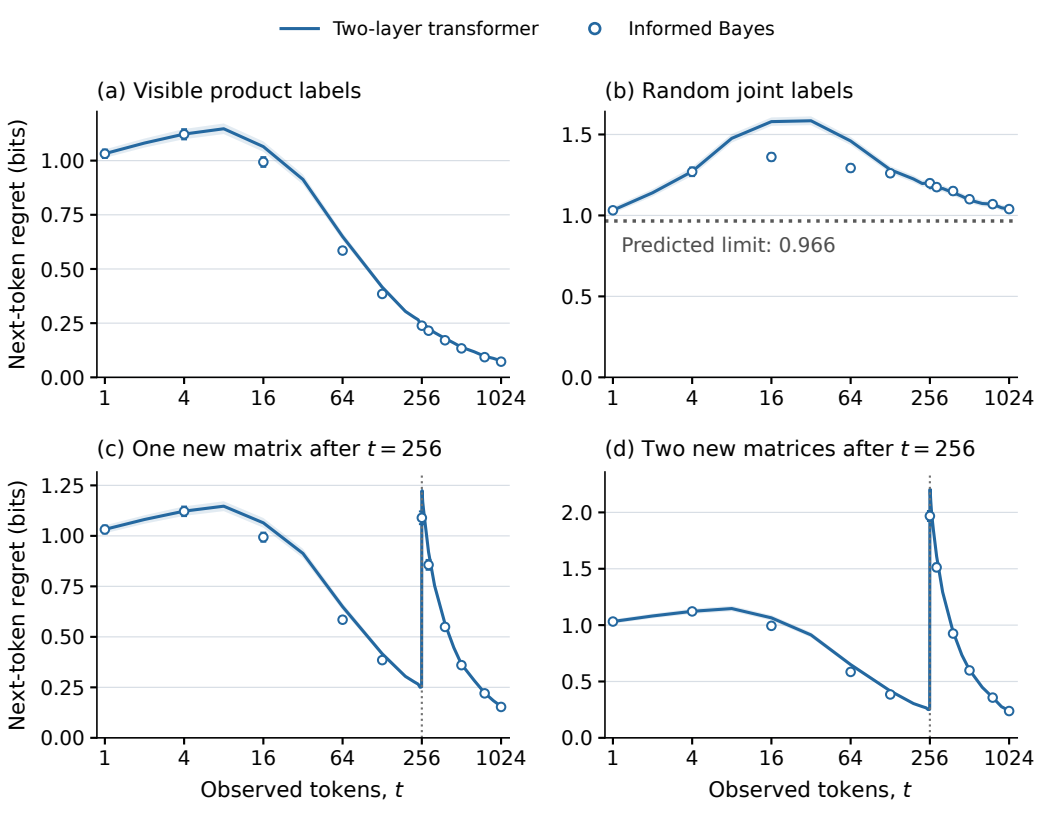
Two-layer transformer
Informed Bayes
(a) Visible product labels
(b) Random joint labels
(c) One new matrix after $t = 256$
(d) Two new matrices after $t = 256$
Next-token regret (bits)
Observed tokens, t
Predicted limit: 0.966
0.00
0.25
0.50
0.75
1.00
1.25
1.50
2.00
1
4
16
64
256
1024